

Bayesian-Based Symptom Screening for Medical Dialogue Diagnosis

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Abstract—In a medical dialogue diagnosis system, the selection of symptoms for inquiry has a significant impact on diagnostic accuracy and dialogue efficiency. In a typical diagnosis process, the symptoms initially reported by users are often insufficient to support an accurate diagnosis, making it necessary to ask users about other symptoms through dialogue to form a conclusive diagnosis. In this paper, we propose a disease diagnosis algorithm based on Bayesian, which simulates the process of doctor's inquiry and diagnosis by dynamically updating the list of diseases to increase the interpretability of diagnosis results. For the symptom interrogation, we propose a symptom screening algorithm based on the difference of symptom sets to exclude diseases with low probability. Through the intersection and union of disease symptom sets, we can screen out the symptoms that can distinguish diseases in fewer inquiring rounds. The experimental results demonstrate the proposed method performs more efficiently than existing state-of-the-art algorithms.

Index Terms—Medical dialogue diagnosis system, symptom screening, Bayesian inference, e-health

I. INTRODUCTION

The medical dialogue diagnosis system is one of the most important artificial intelligence applications in healthcare. The medical dialogue diagnosis system simulates a doctor, inquiring the patient and combining their self-reports to predict potential disease. Medical dialogue diagnosis system can provide pre-consultation to patients and also help doctors to collect information from patients.

The main challenges are that patients describe their symptoms incompletely and that one symptom can be associated with multiple diseases, which makes diagnosis more difficult. The current research interest is to make a more accurate diagnosis by having multiple rounds of dialogue with the patient to make up for information that the patient is not actively giving. Also, using as few dialogue rounds as possible to check enough additional symptoms is desired for the efficiency of this system.

In traditional methods, [1] developed an expert system by extracting knowledge from datasets and human knowledge, but it has poor transferability. Machine learning approaches such as decision tree, SVM, random forest have been used

to predict a disease, do not get good classification results when there is a high overlap of symptoms between diseases. Recently there has been an increase in research based on deep reinforcement learning, [2] defined the query and diagnosis process as Markov decision process and using deep Q-network (DQN) for policy learning. Based on this, [3] introduced knowledge graphs to add knowledge branch and knowledge-routed graph branch, but with limited enhancement. However, RL-based methods have poor interpretability, require more resources, and are also poorly transferable.

In this paper, we propose Bayesian-based symptom screening algorithm. We get the rough probabilities of diseases based on the positive and negative symptoms in the patient's self-report, using Bayesian and ranking them in descending order. Symptom screening combines the variability between disease symptom sets to identify the symptoms that best distinguish diseases, we select high probability symptoms in low probability diseases to get more additional information. Symptom screening is performed using a binary search, which reduces the number of diseases by half in each round of inquiry, greatly improving the efficiency of inquiry. The patient's additional information and self-report are then used to make the final disease prediction. Our contribution is as follows:

1) We propose lightweight Bayesian-based symptom screening algorithm to predict patients' diseases by multiple rounds of interrogation. The main advantages are high interpretability, low computational complexity, and less data resources and computational resources required.

2) We propose symptom screening algorithm. The variability between sets of disease symptoms is exploited to find the symptoms that best distinguish the disease. The number of interrogation rounds is greatly reduced by binary search. The best results are obtained on two real-world datasets.

The rest of this paper is organized as follows. Section II reviews the related works. In Section III, we describes the overall framework and details of the algorithm. We then presents the experimental setup and results in Section IV. Finally, we conclude in Section V.

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II. RELATED WORK

In the work of intelligent diagnosis, the first Bayesian inference and tree-based methods used entropy function to select symptoms based on information gain theory. [4] and [5] proposed to address the cost of feature acquisition by using decision trees and random forest methods. [1] attempted to extract rule-based representations from medical data and human knowledge for diagnosis. [6] and [7] suggested that trained predictive factors for sensitivity analysis to measure the importance of each feature in a given context. [8] developed a hierarchical finding framework based on symptom embedding in which graphical representation learning is customized for disease diagnosis. [9] proposed approach based on the supervised learning of neural models with logic regularization. These methods often use greedy or approximation algorithms due to the global maximization of information gain or the recalcitrance of global sensitivity, resulting in low accuracy.

With the latest progress of deep reinforcement learning and task-oriented dialogue systems, the application research of deep reinforcement learning [10] in medical dialogue systems has gradually increased. In the context of symptom-based dialogue system for disease diagnosis, [2] proposed a human-machine dialogue system for automatic diagnosis. This automatic diagnosis system defined the query and diagnosis process as a Markov decision process, and applied a deep Q-Network (DQN) to learn the policy of the system. To decide whether to continue the inquiry by asking about a symptom or to end the diagnosis by predicting a disease. It establishes a relatively complete dialogue system, which takes the first step to solve the problem of automatic diagnosis of medical dialogue.

KR-DS [3] contains DQN based on knowledge path. By considering the relationship between diseases and symptoms, the knowledge branch and knowledge routing branch are added to the deep reinforcement learning framework, and the external probability symptoms related to the reinforcement learning framework are incorporated into it. It improves the rationality of medical dialogue decision-making.

III. METHODOLOGY

In this section, we introduce the Bayesian-based symptom screening algorithm. The probability distribution of symptoms and diseases in the dataset is used to derive the probability of each disease of a patient by combining the explicit and implicit symptoms. The symptom screening combines the variability between disease symptom sets to identify the symptoms that can best distinguish diseases.

A. Task description

In this work, we consider each patient has only one disease, because it is quite common in some healthcare practice. The symptom status is only negative or positive and the system extracts negative and positive symptoms from the

patient's self-report. Bayesian inference is performed using these symptoms to give the probability of all diseases. The symptom screening algorithm is used during the inquiry stage to complement the patient's symptoms. We assume that the patient will answer the system's questions based on his or her real situation.

B. Notations

The disease set is represented by D , and the size of D is m . The symptom set is represented by S , S represents all symptoms of the disease d , where $d \in D$, S^+ and S^- represent the positive and negative symptom sets, respectively.

C. Bayesian inference for disease diagnosis

Based on the distribution of the datasets, we can get the probability $P(d = 1)$ of each disease being positive and the probability $P(s = 1|d = 1)$ of each symptom being positive under each disease being positive, where $s \in S$. Given the symptom status information, the conditional probability of each disease can be obtained as follows:

$$P(d = 1|s = 1) = \frac{P(d = 1, s = 1)}{P(s = 1)} \quad (1)$$

where

$$P(s = 1) = \sum_{d \in D} P(s = 1|d = 1) * P(d = 1) \quad (2)$$

We cannot confirm that the disease d is negative when the symptom s is negative, so the probability that both are negative can be

$$\begin{aligned} P(d = 0|s = 0) \\ &= 1 - P(d = 1|s = 0) \\ &= 1 - \frac{\sum_{d \in D} P(s = 0|d = 1) * P(d = 1)}{P(s = 0)} \end{aligned} \quad (3)$$

where

$$P(s = 0) = 1 - P(s = 1) \quad (4)$$

according to equation 1-4 we can get

$$P(d = 1|S^+, S^-) = \frac{P(d = 1, S^+, S^-)}{P(S^+, S^-)} \quad (5)$$

where

$$\begin{aligned} P(d = 1, S^+, S^-) \\ &= P(d = 1) \prod_{s \in S^+} P(s = 1|d = 1) \prod_{s \in S^-} P(s = 0|d = 1) \\ &= P(d = 1) \prod_{s \in S^+} P(s = 1|d = 1) \prod_{s \in S^-} (1 - P(s = 1|d = 1)) \end{aligned} \quad (6)$$

$$P(S^+, S^-) = \sum_{d \in D} P(d = 1, S^+, S^-) \quad (7)$$

Note that the term $P(d = 1, S^+, S^-)$ can be factorized because the symptom are conditionally independent given all diseases.

The Bayesian inference algorithm is summarized as Algorithm 1. Step 1 initializes an empty list D_P to store the probabilities of all diseases. Step 2 calculates for each disease, the joint probability of S^+ and S^- . Step 3 calculates the probability of S^+ and S^- . Step 4 calculates the conditional probabilities of each disease when the conditions of S^+ and S^- are all added to the D_P .

Algorithm 1: Bayesian inference algorithm for diagnosis

Input: Positive symptom S^+ and negative symptom S^- .

Output: List D_P of probability in each disease

```

1 Initialize an empty list  $D_P$  ;
2 for  $i$  in  $1...m$  do
3   | Compute  $P(d_i = 1, S^+, S^-)$  using equation 6;
4 end
5 Compute  $P(S^+, S^-)$  using equation 7;
6 for  $i$  in  $1...m$  do
7   | Compute  $P(d_i = 1|S^+, S^-)$  using equation 5;
8   | add it to the list  $D_P$ ;
9 end
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D. Symptom screening

Generally, patient do not provide enough information for the doctor to confirm the disease, so additional information is required through multiple rounds of inquiry. We are inspired by this to propose symptom screening algorithm based on the difference of symptom sets of each disease.

Patient self-reports were used to get rough disease probabilities using Algorithm 1 and sorted in descending order (eq $P(d_0|S^+, S^-) > P(d_1|S^+, S^-) > \dots > P(d_{m-1}|S^+, S^-)$). Then we get the high probability disease symptom set $S_H = \{S_i, i \in [1, 2, \dots, \lceil \frac{m}{2} \rceil]\}$ and the low probability disease symptom set $S_L = \{S_i, i \in [\lceil \frac{m}{2} \rceil + 1, \dots, m]\}$. Figure 1 shows

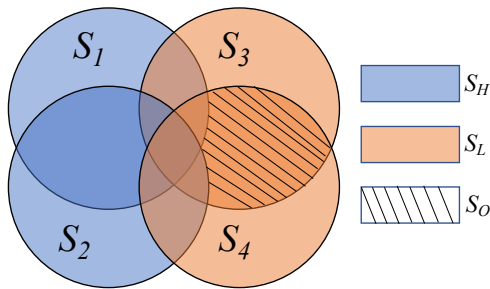


Fig. 1. A round of symptom screening process

our symptom screening process for each round, selected to

take situations with four diseases. Each circle represents the full set of symptoms of a disease, the slash area is the target symptom set, blue and orange represent the symptom sets belonging to S_H and S_L , respectively. Our goal was to find the symptom that best distinguished the disease. We put the symptom into S^+ or S^- depending on the patient's negative or positive response at each inquiry, while removing half of the symptom set. Because the current results are biased towards high probability diseases, the system receives less benefit if the symptoms are selected from the high probability disease symptom set to ask the patient, in other words, the Bayesian inference process gets more additional information when the symptoms are selected from the low probability disease symptom set to ask the patient.

In order to make the screened symptoms correlated with all sets in S_L and uncorrelated with all sets in S_H , we take the union set of S_i in S_H :

$$S_U = \bigcup_{S_i \in S_H} S_i \quad (8)$$

we take the intersect set of S_i in S_L :

$$S_\cap = \bigcap_{S_i \in S_L} S_i \quad (9)$$

Then, we get the target set of symptoms:

$$S_O = S_\cap - S_U = \bigcap_{S_i \in S_L} S_i - \bigcup_{S_i \in S_H} S_i \quad (10)$$

After we get the target symptom set S_O , in order to make the selected symptoms work in most cases, so we select the most frequently occurring symptoms according to the probability distribution of the symptoms, which is the following:

$$s_{Omax} = \underset{s \in S_O}{\operatorname{argmax}} P(s = 1) \quad (11)$$

So we select the symptom s_{Omax} with the highest probability of positive in the S_O , add it to the positive symptom set S^+ or negative symptom set S^- according to the patient's answer.

Figure 2 represents the complete interrogation process of the system when four diseases are present. The dashed circles indicate the set of eliminated symptoms. In the first round, the system gets the current s_{Omax} symptoms by the symptom screening algorithm and interrogates the patient. When s_{Omax} is positive, S_1 and S_2 are eliminated and s_{Omax} is put into the positive symptom set S^+ ; when the interrogation result of s_{Omax} is negative, S_3 and S_4 are eliminated and s_{Omax} is put into the negative symptom set S^- . By the second round, take the left half as an example. At this point, s_{Omax} is recalculated, and when s_{Omax} is positive, S_3 is eliminated and s_{Omax} is put into the positive symptom set S^+ ; the current when s_{Omax} is negative, S_4 is eliminated and s_{Omax} is put into the negative symptom set S^- . After the second round, only one disease symptom

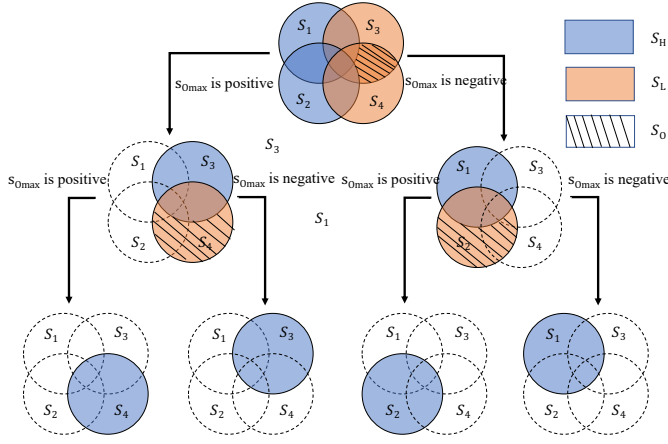


Fig. 2. The symptom screening process in the four disease symptom sets

Algorithm 2: Symptom screening algorithm based on Bayesian inference

Input: Positive symptom S^+ and negative symptom S^- , maximum step N .

Output: pick the disease with the highest probability in the list D_P

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1   $n = 0$ ;
2  Compute list  $D_P$  of probability in each disease using
   Algorithm 1;
3  while  $n \leq N$  do
4      Calculate the length  $L$  of  $D_P$ ;
5      if  $L \leq 1$  or  $n = N$  then
6          Compute list  $D_P$  of probability in each
           disease using Algorithm 1;
7          return
8      else
9           $n = n + 1$ ;
10         Sort the disease probabilities in  $D_P$  in
           descending order;
11         Get  $S_H$  and  $S_L$  according to  $D_P$ ;
12         Get  $S_O$  using equation 10;
13         Get  $s_{Omax}$  using equation 11;
14         if  $s_{Omax}$  is positive then
15             Add  $s_{Omax}$  to the positive symptom set
               $S^+$ ;
16              $D_P \leftarrow D_P[\lceil \frac{L}{2} \rceil + 1, \dots, L]$ ;
17         else
18             Add  $s_{Omax}$  to the negative symptom set
               $S^-$ ;
19              $D_P \leftarrow D_P[1, \dots, \lceil \frac{L}{2} \rceil]$ ;
20         end
21     end
22 end

```

set remains, and this can be stopped. The final disease probabilities are obtained by putting the updated S^+ and S^- using Algorithm 1.

The symptom screening is summarized as Algorithm 2. Step 1 obtains the list of disease probabilities D_P based on Algorithm 1 and S^+ and S^- in the patient's self-report. Step 2 calculates the length of the current disease list during each interrogation, arranges the diseases of D_P in descending order of probability, and obtains s_{Omax} after the symptom screening algorithm. Step 3 interrogates the patient, adds s_{Omax} to S^+ if it is positive, and updates the disease probability list to low-probability diseases. If negative, the s_{Omax} is added to S^- and the disease probability list is updated to high probability disease. When there is only one disease left in the disease list, use Algorithm 1 to output the disease with the highest probability.

IV. EXPERIMENT

A. Dataset

MZ [2] was collected on the China Baidu Thumb Doctor online consultation platform in 2018. It collected four common childhood diseases and contains 710 conversations covering 66 symptoms of 4 diseases. The corresponding relationships between the four diseases and the number of conversations and symptoms are shown in Table I.

DX [3] was collected from an online healthcare community in China (Lilac Garden), which retained the original self-report and interaction discourse. DX annotation for five types of diseases, including allergic rhinitis, upper respiratory tract infection, pneumonia, hand, foot and mouth disease in children and childhood diarrhea, [3] took in self-reported symptoms, and the conversation and normalized to 41 kinds of symptoms, shown in Table I.

TABLE I
THE NUMBER OF CONVERSATIONS AND SYMPTOMS OF EACH DISEASE

dataset	disease	dialogue	symptoms
MZ	pediatric diarrhea	200	56
	upper respiratory tract infection	160	59
	minor bronchitis	200	61
	pediatric dyspepsia	150	66
DX	allergic rhinitis	102	24
	upper respiratory tract infections	122	23
	pneumonia	100	29
	Hand, foot and mouth disease	101	22
	childhood diarrhea	102	33

B. Baseline

SVM treats disease diagnosis as a multi-class classification problem. It takes as input a one-hot vector representation of the patient's symptoms and predicts disease. There are two configurations: one that takes both explicit and implicit symptoms as inputs (denoted SVM-ex+im), and the other

TABLE II
PERFORMANCE COMPARISONS ON MZ DATASET.

method	upper respiratory tract infection	minor bronchitis	pediatric diarrhea	pediatric dyspepsia	Overall	Ave rounds
SVM-ex	0.44	0.71	0.89	0.28	0.59	-
SVM-ex+im	0.52	0.93	0.91	0.34	0.71	-
Basic DQN	-	-	-	-	0.65	-
KR-DS	-	-	-	-	0.73	-
Bayesian-ex	0.5333	0.8235	0.77	0.3733	0.6338	1
Bayesian-ex+im	0.5667	0.8235	0.8444	0.3939	0.6761	3.26
ours	0.833	0.9412	0.9778	0.909	0.9225	2.89

TABLE III
PERFORMANCE COMPARISONS ON DX DATASET.

method	allergic rhinitis	upper respiratory tract infections	pneumonia	Hand, foot and mouth disease	childhood diarrhea	Overall	Ave rounds
Basic DQN	-	-	-	-	-	0.731	3.92
KR-DS	-	-	-	-	-	0.74	3.36
Bayesian-ex	0.6	0.625	0.55	0.9	1	0.7308	1
Bayesian-ex+im	0.95	0.3750	0.85	0.85	0.95	0.7788	2.67
ours	1	0.9167	0.85	1	1	0.9519	3.08

that uses only explicit symptoms to predict disease (denoted SVM-ex).

Basic DQN defines the query and diagnosis process as a Markov decision process, and on this basis, deep Q-network (DQN) is applied to learn the system policy-wise to decide whether to continue the conversation by asking about a symptom or to end the diagnosis by predicting a disease.

KR-DS adds knowledge branches and knowledge routing branches to the deep reinforcement learning framework by considering the relationship between diseases and symptoms. It improves the rationality of medical dialogue decision-making.

C. Setup

In terms of data set division, according to the division of data set by [2] and [3], training set: test set = 568:142 in MZ data set. In the DX dataset, training set: test set = 423:104. Symptoms with "UNK" symptom value were identified as invalid symptom information and the symptom was eliminated. And only when the disease and symptom are both positive, the symptom can be added to the disease symptom set. The initial inquiry round number turn is 1, because after the user enters the user report, a full inquiry is conducted even if the doctor does not ask about other symptoms directly given the diagnosis. Subsequently, when the system asks the user for symptoms, the dialogue rounds is increased by 1. Our evaluation methods are 1) diagnostic accuracy, where the diagnosis succeeds when the predicted result agrees with the true label, and 2) average dialogue rounds, where the system gets the average dialogue rounds needed for each patient's disease result.

D. Result

As shown in Tables II and III, the accuracy of Bayesian for most diseases in the MZ dataset is higher than SVM. When only explicit symptoms are used for disease diagnosis, Bayesian outperforms SVM in the classification of upper respiratory tract infection, pediatric bronchitis, and pediatric dyspepsia by approximately 9%. Additionally, the overall accuracy of Bayesian increased by 4.38%. However, when both explicit and implicit symptom information are used for diagnosis, the accuracy of Bayesian was 3.39% lower than that of SVM, which showed a 22% improvement in disease diagnosis accuracy. The experimental results proved that Bayesian was better than that of SVM when only explicit symptoms were used for diagnosis. However, Bayesian was similar to that of SVM when the disease was diagnosed according to both explicit and implicit symptom information.

Both Basic DQN and KR-DS are based on the explicit symptoms in the data set, and by learning the symptom rules asked by doctors in the data set, they can screen symptoms and make decisions. As shown in table III and II, compared with the best results, the disease diagnosis algorithm is improved by 19.25% and 21.19%, respectively.

In the MZ dataset, the average dialogue rounds in the test set is 4.20, and the average dialogue rounds of the disease diagnosis algorithm proposed in this paper is 2.89, which is 1.31 rounds less (31.19% decrease) compared with the average dialogue rounds in the dataset, and the improvement of the disease prediction accuracy is 24.64%. When adding constraints to the symptom screening of the disease diagnosis algorithm to limit the dialogue rounds to no more than the rounds in the test set, the dialogue rounds of the disease diagnosis algorithm is 2.65, and the accuracy of the disease diagnosis algorithm is 85.91%, which is 6.34% lower than

the initial constraint, but the overall accuracy of the disease diagnosis algorithm is still 18.30% improvement.

In the DX dataset, the average dialogue rounds in the test set was 2.76, and the average dialogue rounds for the disease diagnosis algorithm in this paper was 3.04, an increase of 0.28 rounds over the average inquiring rounds in the dataset, but a decrease of 0.32 (9.5%) over the inquiring rounds in KR-DS. Adding constraints to the symptom screening module of the disease diagnosis algorithm to limit it to no more than the rounds in the dataset, the diagnostic accuracy of the disease diagnosis algorithm was 90.38%, at which point it improved by 12.50% and 16.38% compared to the results of Bayesian-ex+im and KR-DS, respectively.

V. CONCLUSION

In this paper, we propose disease diagnosis algorithm based on Bayesian and a symptom screening algorithm based on the difference of disease symptom set to solve the problems of low diagnostic accuracy, long running time and poor interpretability of existing reinforcement learning models in this scenario. The algorithm achieved the best results on two real-world datasets.

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